Random Variable and Probability Distribution

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Outline

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2. Probability Distribution
3. Cumulative Distribution Function (CDF)
4. Mathematical Expectation and Variance

Random Variable

Introduction

A variable which takes numerical values resulting from random experiments, is called a random variable (RV).

- There is a probability associated with each possible values
- Random variables are denoted by capital letters such as X, Y, Z etc.
- Possible values are denoted by small letters such as x, y, z etc.
- Example:
  - Height of students
  - Number of heads when tossing a coin three times

Types of Random Variable

- **Discrete random variable**: A random variable defined over a discrete sample space
  - Number of students in a class, sample space, $S = \{0, 1, 2, ..., \infty\}$
  - Number of correct answers in among 50 questions, $S = \{0, 1, 2, ..., 50\}$
- **Continuous random variable**: A random variable defined over a continuous sample space
  - Monthly income, $S = \{x : 0 \leqslant x < \infty\}$
  - Monthly profit, $S = \{x : -\infty < x < \infty\}$

Probability Distribution

Probability Distribution

- Probability distribution means the distribution of the probabilities among the different values of a random variable
- For example, when tossing a coin twice, the sample space, $S = \{HH, HT, TH, TT\}$
- Let an RV, $X =$ number of heads in two coin tosses, then X can take the values: $0, 1, 2$
- Then the probability of each value of X:

x	$P(X = x)$
0	$1/4$
1	$2/4$
2	$1/4$

Types of Probability Distributions

Depending on the type of variable, distributions are of two types:

- **Discrete probability distribution**: probability distribution of a discrete random variable
- **Continuous probability distribution**: probability distribution of a continuous random variable

Discrete probability distribution

- The probability distribution of a discrete random variable is a table, graph, formula, or other device used to specify all possible values of a discrete random variable along with their respective probabilities
- If we let the discrete probability distribution be represented by the function $p(x)$, then $p(x) = P(X = x)$ is the probability of the discrete random variable X to assume a value x
- $p(x)$ is called a probability mass function (PMF)

Probability Mass Function (PMF)

A probability mass function gives the probability of each of the value of a discrete random variable.

A function, $p(x)$, of a discrete random variable X will be called a PMF, if and only if all the following conditions are satisfied:

1. $p(x) \geq 0; \ \forall x$
2. $\sum_{x} p(x) = 1$

The probability for a value a of the random variable X is written as $p(a)$, i.e., $p(a) = P(X = a)$.

Example: PMF

In page 3, the probability mass function of the variable X representing the number of heads when a coin is tossed twice is given. It is an example of a PMF.

x	$P(X = x)$
0	$1/4$
1	$2/4$
2	$1/4$

Probabiilty Density Function (PDF)

The probability distribution of a continuous random variable is called continuous probability distribution. The probabilities are given by the probability density function (PDF) of that distribution.

A function $f(x)$ of a continuous random variable X is called the PDF of X if and only if the following conditions are satisfied:

1. $f(x) \geqslant 0; \forall x$
2. $\int_x f(x)dx = 1$

   Here, $\int_x f(x)dx$ means integration of $f(x)$ over the whole sample space of X. Sometimes, it's also written as $\int_{-\infty}^{\infty} f(x)dx$.

In the continuous case, the probability that $X \in [a, b]$ is found by integrating the PDF as: $P(a \leqslant X \leqslant b) = \int_a^b f(x)dx$.

PDF: Example

Suppose X is a continuous random variable whose density function is given by:

$$f(x) = \begin{cases} C(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

1. What is the value of C?
2. Find $P(X > 1)$.

PDF: Example Solution

1.

$$
\begin{aligned}
&\int_x f(x)dx = 1 \\
\Rightarrow &\int_0^2 C(4x - 2x^2)dx = 1 \\
\Rightarrow &C\left[4x^2/2 - 2x^3/3\right]_0^2 = 1 \\
\Rightarrow &C = 3/8
\end{aligned}
$$

2.

$$P(X > 1) = \int_1^2 \frac{3}{8}(4x - 2x^2) = 1/2$$

Probability at a Single Point for Continuous Random Variables

For a continuous RV, probability over an interval is:

$$P(a \leqslant X \leqslant b) = \int_a^b f(x)\,dx$$

At a single point, the interval collapses to zero width, so:

$$P(X = a) = \int_a^a f(x)\,dx = 0$$

Since $P(X = a) = 0$, adding or removing a boundary point never changes probability:

- $P(X \leqslant a) = P(X < a)$ and $P(X \geqslant a) = P(X > a)$
- $P(a \leqslant X \leqslant b) = P(a < X < b)$

Cumulative Distribution Function (CDF)

Cumulative Distribution Function (CDF)

The cumulative distribution function (CDF) gives the probability that a random variable X is lower than some value x. Sometimes, it is also called "distribution function". It is denoted by $F(x)$.

- For discrete random variable, $F(x) = P(X \leqslant x) = \displaystyle\sum_{k=-\infty}^{x} p(k)$
- For continuous case, $F(x) = P(X < x) = P(X \leqslant x) = \displaystyle\int_{-\infty}^{x} f(x)dx$
- It is a non-decreasing function, the largest value $F(x)$ can take is 1

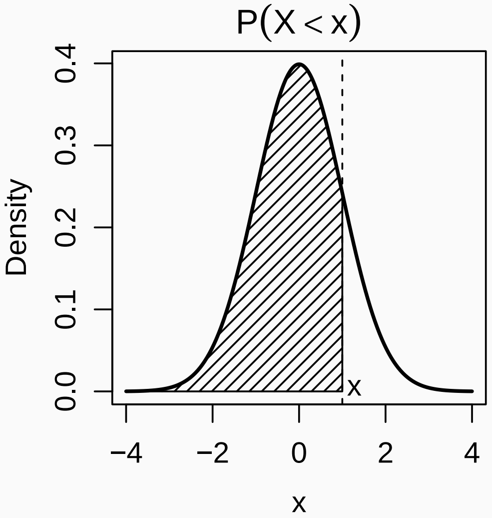
Here, summing/integrating from $-\infty$ means that the lower limit of the sum/integral is the smallest value in the domain of the random variable X.

Graphical Representation of CDF

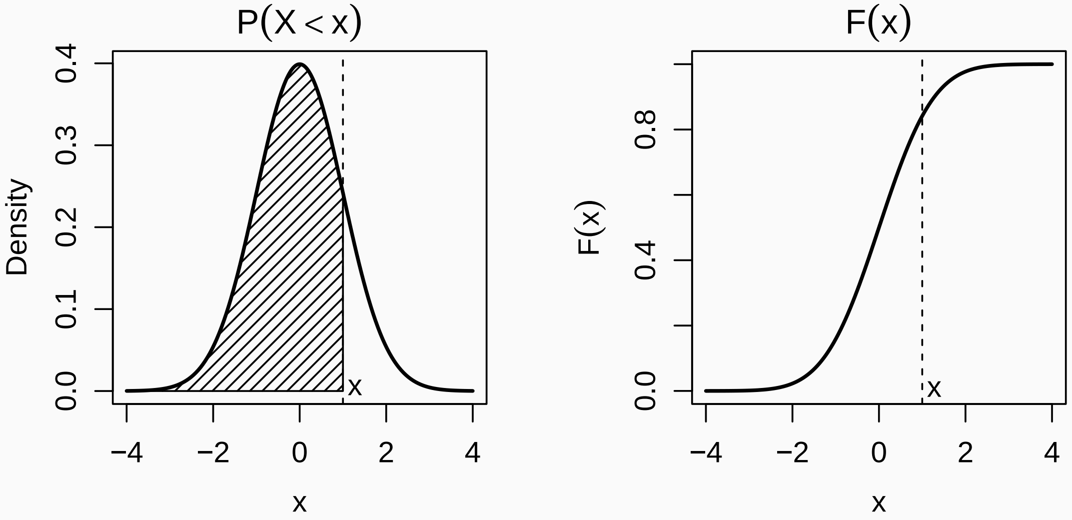
In the plot, X is a continuous random variable, it is "normally" distributed (bell shaped curve).

The shaded region highlights the probability that the random variable X is less than some value x (in the plot, this value is shown to be 1 as an example).

Therefore, the highlighted area is the probability that X is less than 1.



Graphical Representation of CDF (cont.)



CDF: Example 1

In page 3, the probability mass function of the variable X representing the number of heads when a coin is tossed twice is given.
Below, its CDF is also given:

x	$P(X = x)$	$P(X \leq x)$
0	$1/4$	$1/4$
1	$2/4$	$3/4$
2	$1/4$	1

CDF: Example 2

Suppose X is a continuous random variable whose density function is given by:

$$f(x) = \begin{cases} \frac{3}{8}(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find the CDF of X:

$$\begin{aligned} F(x) &= \int_0^x f(x)dx \\ &= \int_0^x \frac{3}{8}(4x - 2x^2)dx \\ &= \frac{3}{8}[(4x^2/2 - 2x^3/3)]_0^x = \frac{3}{8}[(2x^2 - 2x^3/3) - 0] = \frac{3}{8}(2x^2 - 2x^3/3) \end{aligned}$$

Finding Probability from CDF

Continuous RV:

- $P(X \leqslant a) = F(a)$
- $P(X > a) = 1 - F(a)$
- $P(a < X < b) = F(b) - F(a)$

Since $P(X = a) = 0$, the $\leqslant$ and $<$ signs are interchangeable for continuous RVs.

Discrete RV:

- $P(X \leqslant a) = F(a)$
- $P(X < a) = F(a) - P(X = a)$
- $P(X > a) = 1 - F(a)$
- $P(a < X \leqslant b) = F(b) - F(a)$
- $P(a \leqslant X \leqslant b) = F(b) - F(a) + P(X = a)$

Example: Finding Probability from CDF of a Continuous RV

In page 16, we found that for a continuous RV X with PDF given by $f(x) = \frac{3}{8}(4x - 2x^2)$ for $0 < x < 2$, the CDF is $F(x) = \frac{3}{8}(2x^2 - \frac{2x^3}{3})$.

Find $P(0.5 < X < 1.5)$ and $P(0.5 \leqslant X \leqslant 1.5)$:

Using the CDF rule:

$$\begin{aligned} P(0.5 < X < 1.5) = F(1.5) - F(0.5) &= 0.84375 - 0.15625 \\ &= 0.6875. \end{aligned}$$

Since X is continuous, $P(X = a) = 0$.

Therefore, $P(0.5 \leqslant X \leqslant 1.5) = 0.6875$.

Example: Finding Probability from CDF of a Discrete RV

Suppose the CDF of a random variable X satisfies: $F(2) = 0.35$, $F(5) = 0.80$ and $P(X = 2) = 0.10$. Find $P(2 < X \leqslant 5)$ and $P(2 \leqslant X \leqslant 5)$.

First,

$$\begin{aligned} P(2 < X \leqslant 5) &= F(5) - F(2) \\ &= 0.80 - 0.35 = 0.45. \end{aligned}$$

Next,

$$\begin{aligned} P(2 \leqslant X \leqslant 5) &= F(5) - F(2) + P(X = 2) \\ &= 0.45 + 0.10 = 0.55. \end{aligned}$$

Mathematical Expectation and Variance

Mathematical Expectation

Expectation of an RV, X is the weighted average of the possible values of X. The weight of each value of X is its corresponding probability.

- When X is discrete: $\mathrm{E}(X) = \sum_x x \cdot p(x)$
- When continuous: $\mathrm{E}(X) = \int_x x \cdot f(x) dx$
- The sum/integral is over all values of X
- It is often denoted by μ, that is: $\mathrm{E}(X) = \mu$
- It is also known as expected value or population mean

Variance of a Random Variable

The variance of an RV, X measures the dispersion in the values of X. It is denoted by $\mathrm{Var}(X)$ or $\mathrm{V}(X)$.

$$\begin{aligned}
\mathrm{V}(X) &= \mathrm{E}[(X - \mathrm{E}(X))^2] \\
&= \mathrm{E}[X^2 - 2X \cdot \mathrm{E}(X) + [\mathrm{E}(X)]^2] \\
&= \mathrm{E}(X^2) - 2E(X) \cdot \mathrm{E}(X) + [\mathrm{E}(X)]^2 \\
&= \mathrm{E}(X^2) - 2[\mathrm{E}(X)]^2 + [\mathrm{E}(X)]^2 \\
&= \mathrm{E}(X^2) - [\mathrm{E}(X)]^2
\end{aligned}$$

Here, $\mathrm{E}(X^2)$ is given by $\sum_x x^2 \cdot p(x)$ or $\int_x x^2 \cdot f(x)\,dx$ for discrete and continuous case respectively.

Expectation and Variance: Example 1

In page 3, the PMF of the random variable X representing the number of heads in two throws of a fair coin is given.
X can take the values 0, 1 and 2 with probabilities $1/4$, $1/2$ and $1/4$ respectively.
The expected value of X is:

$$\mathrm{E}(X) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 0 + \frac{1}{2} + \frac{1}{2} = 1$$

The variance of X is:

$$\begin{aligned} \mathrm{V}(X) &= \mathrm{E}(X^2) - [\mathrm{E}(X)]^2 \\ &= [0^2 \cdot \frac{1}{4} + 1^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{4}] - 1^2 \\ &= [0 + \frac{1}{2} + 1] - 1 = \frac{1}{2} \end{aligned}$$

Expectation and Variance: Example 2

Find the expected value and the variance of the RV X whose PDF is given in page 16.

$$\begin{aligned}
\mathrm{E}(X) &= \int_0^2 x \cdot \frac{3}{8}(4x - 2x^2)dx \\
&= \frac{3}{8}\int_0^2 (4x^2 - 2x^3)dx \\
&= \frac{3}{8}\left[4x^3/3 - 2x^4/4\right]_0^2 \\
&= \frac{3}{8}\left[4 \times 2^3/3 - 2 \times 2^4/4 - 0\right] = 1
\end{aligned}$$

Finding the variance is left as an exercise.

Properties of Expectation and Variance

Let a, b and c be some constants, and X and Y be two independent random variables.

1. $\mathrm{E}(c) = c$
2. $\mathrm{E}(cX) = c\,\mathrm{E}(X)$
3. $\mathrm{E}(c + X) = c + \mathrm{E}(X)$
4. $\mathrm{E}(aX + bY) = a\,\mathrm{E}(X) + b\,\mathrm{E}(Y)$

1. $\mathrm{V}(c) = 0$
2. $\mathrm{V}(cX) = c^2\,\mathrm{V}(X)$
3. $\mathrm{V}(c + X) = \mathrm{V}(X)$
4. $\mathrm{V}(aX + bY) = a^2\,\mathrm{V}(X) + b^2\,\mathrm{V}(Y)$

If X and Y are dependent:
$\mathrm{V}(aX + bY) = a^2\,\mathrm{V}(X) + b^2\,\mathrm{V}(Y) + 2ab\,\mathrm{CoV}(X, Y)$.

Here, $\mathrm{CoV}(X, Y)$ is the covariance of X and Y. More on covariance in a future lecture.

Properties of Expectation and Variance: Example 1

Suppose X is a random variable with $\mathrm{E}(X) = 5$ and $\mathrm{V}(X) = 2$.

- $\mathrm{E}(2X) = 2\,\mathrm{E}(X) = 2 \times 5 = 10$
- $\mathrm{V}(2X) = 2^2\,\mathrm{V}(X) = 4 \times 2 = 8$
- $\mathrm{E}(2X - 3) = 2\,\mathrm{E}(X) - 3 = 2 \times 5 - 3 = 10 - 3 = 7$
- $\mathrm{V}(2X - 3) = 2^2\,\mathrm{V}(X) = 4 \times 2 = 8$

Properties of Expectation and Variance: Example 2

Suppose X and Y are two random variables with $\mathrm{E}(X) = 5$, $\mathrm{V}(X) = 2$, $\mathrm{E}(Y) = 3$ and $\mathrm{V}(Y) = 4$.

If X and Y are independent:

- $\mathrm{E}(X + Y) = \mathrm{E}(X) + \mathrm{E}(Y) = 5 + 3 = 8$
- $\mathrm{V}(X + Y) = \mathrm{V}(X) + \mathrm{V}(Y) = 2 + 4 = 6$
- $\mathrm{E}(2X - 3Y) = 2\,\mathrm{E}(X) - 3\,\mathrm{E}(Y) = 10 - 9 = 1$
- $\mathrm{V}(2X - 3Y) = 2^2\,\mathrm{V}(X) + 3^2\,\mathrm{V}(Y) = 8 + 36 = 44$

If instead X and Y are dependent, with $\mathrm{CoV}(X, Y) = 1$:

- $\mathrm{V}(2X - 3Y) = 2^2\,\mathrm{V}(X) + 3^2\,\mathrm{V}(Y) + 2(2)(-3)\,\mathrm{CoV}(X, Y) = 8 + 36 - 12 = 32$

Exercises

1. A discrete random variable X has PMF $p(x) = k(5 - x)$ for $x = 1, 2, 3, 4$, and $p(x) = 0$ otherwise. Find the value of k, then find $P(X \geqslant 3)$.
2. A continuous random variable X has PDF $f(x) = kx$ for $0 < x < 4$, and $f(x) = 0$ otherwise. Find the value of k, then find $P(X > 2)$.
3. A discrete random variable X has PMF $P(X = 1) = 0.2$, $P(X = 2) = 0.3$ and $P(X = 3) = 0.5$. Find the CDF $F(x)$ of X.
4. A continuous random variable X has PDF $f(x) = 2x$ for $0 < x < 1$, and $f(x) = 0$ otherwise. Find the CDF $F(x)$ of X.

Exercises (cont.)

5. Suppose the CDF of a discrete random variable X satisfies $F(3) = 0.45$, $F(7) = 0.85$ and $P(X = 3) = 0.15$. Find $P(3 < X \leqslant 7)$ and $P(3 \leqslant X \leqslant 7)$.
6. Suppose the CDF of a continuous random variable X is $F(x) = x^2$ for $0 \leqslant x \leqslant 1$. Find $P(0.3 < X < 0.7)$.
7. A discrete random variable X has PMF $P(X = 0) = 0.3$, $P(X = 1) = 0.4$ and $P(X = 2) = 0.3$. Find $E(X)$ and $V(X)$.
8. A continuous random variable X has PDF $f(x) = 3x^2$ for $0 < x < 1$, and $f(x) = 0$ otherwise. Find $E(X)$ and $V(X)$.

Thank you.